Transformers: 2017 vs 2026
(Multimodal LLMs, MoE, Attention at Scale)

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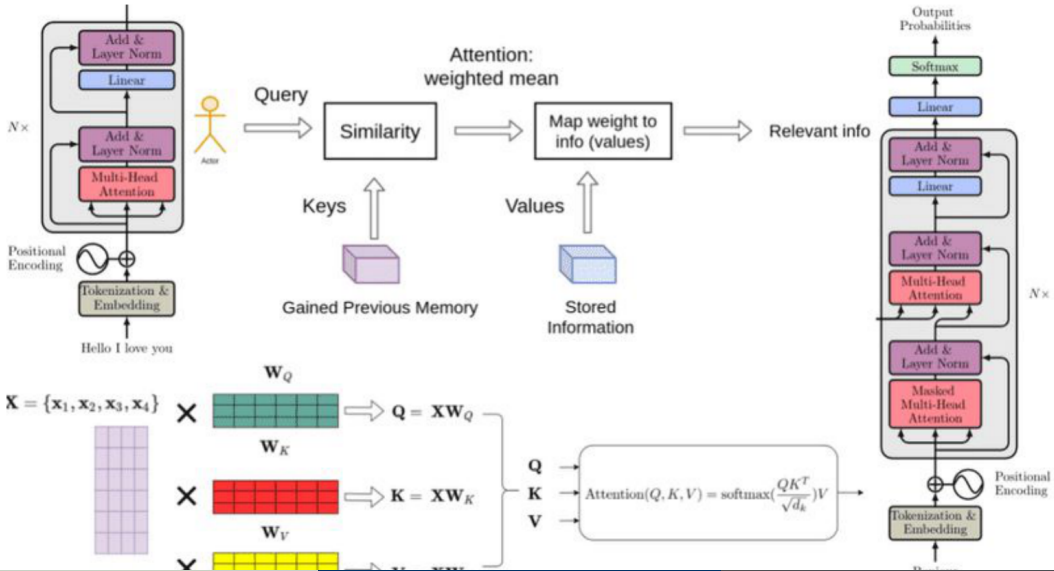
Four Key Engineering Changes (2017 → 2026)

1. **Bandwidth matters more than just speed:** New attention methods like FlashAttention focus on moving data efficiently and using memory well (built-in memory or RAG based retrieval).
2. **From fixed to flexible compute:** Models now adjust their computation and memory use based on the input (using tools like RoPE, sparse MoE, and adaptive resolution).
3. **From separate to unified input streams:** Instead of having different pieces for text, images, etc., everything is processed together in one sequence—with new practical challenges.
4. **More thinking during use:** Some new models use extra internal steps ("thinking tokens") for harder questions, shifting cost to longer decoding instead of just bigger models.

The 2017 Transformer



Why the Transformer?

- The original 2017 Transformer was designed to address key limitations of Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) units:
  - Lack of parallelization
  - Vanishing gradient problem inherent in sequential processing
- Replaced recurrence with self-attention to allow all tokens in a sequence to be processed simultaneously during training[1]
- Enabled significant reductions in wall-clock time needed for model convergence

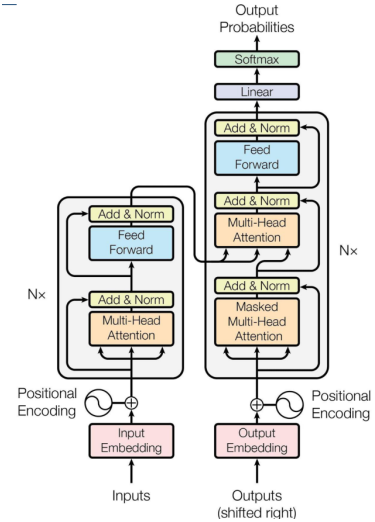
[1] See Vaswani et al., "Attention is All You Need," 2017.

Introduction to Transformers

- **What is a Transformer?**
  - Introduced by Vaswani et al. (2017) – "Attention is All You Need"
- **Key features:**
  - No recurrence, all attention
  - Encoder-decoder structure
  - Enables parallel processing
  - Scales well with data and compute

Transformer Architecture: Components Overview

- Tokenization
- Input Embeddings
- Position Encodings
- Query, Key, & Value
- Attention
- Self Attention
- Multi-Head Attention
- Feed Forward
- Add & Norm
- Encoders
- Masked Attention
- Encoder Decoder Attention
- Linear
- Softmax
- Decoders
- Encoder-Decoder Models



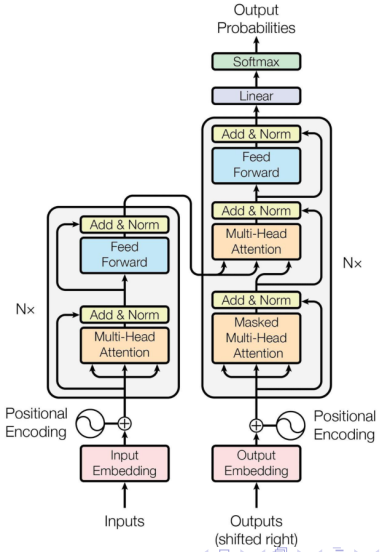
Machine Translation

Targets

Ich habe einen
Apfel gegessen

Inputs

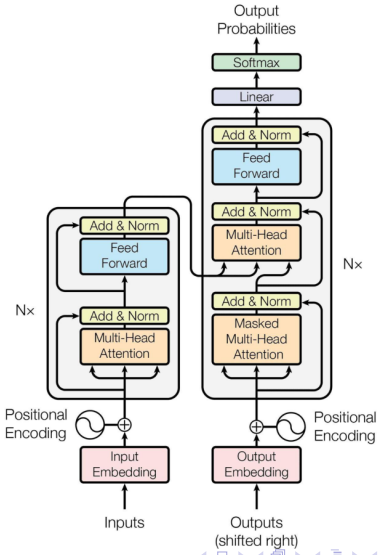
I ate an apple



Processing Inputs

Inputs

I ate an apple



Self-Attention Mechanism

- Computes contextual representation of a word using all other words
- **Inputs:** Query (Q), Key (K), Value (V)
- **Output:** Weighted sum of values

Attention formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

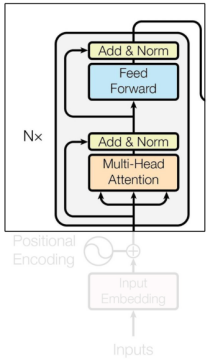
- Enables each word to attend to all others

Self-Attention Intuition: The "Bank" Analogy

- **Query:** What we're looking for (e.g., the meaning of "bank" in context)
- **Keys:** Tags or features associated with each word/sentence (e.g., "river", "money")
- **Values:** Information linked to each key (e.g., context-specific meaning)
- Uses dot-product similarity between Query and Keys to assign attention weights
- Weighted sum of Values produces the contextual representation

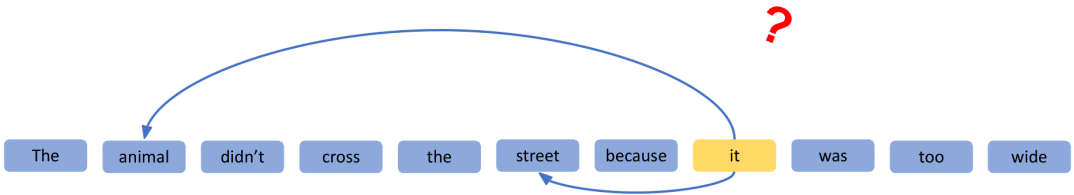
Self Attention

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$



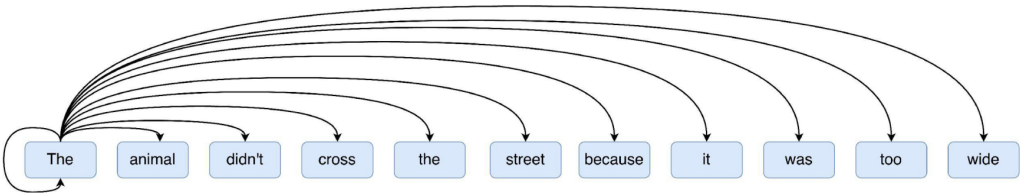
Self Attention

Self Attention

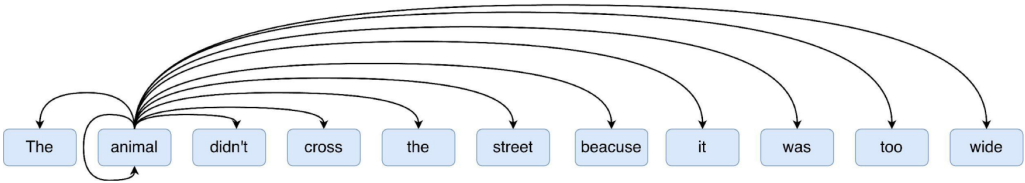


coreference resolution?

Self Attention



Self Attention



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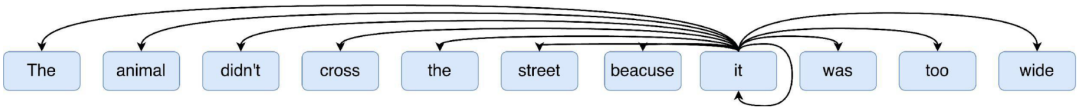
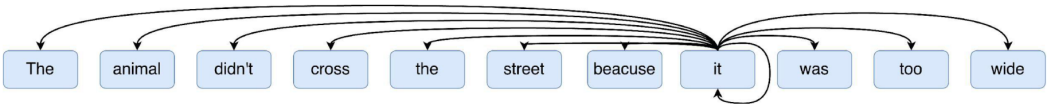


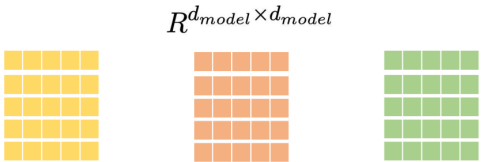
Self Attention

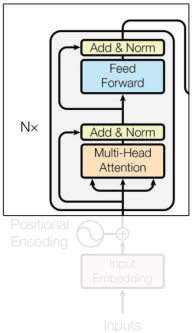


SELF

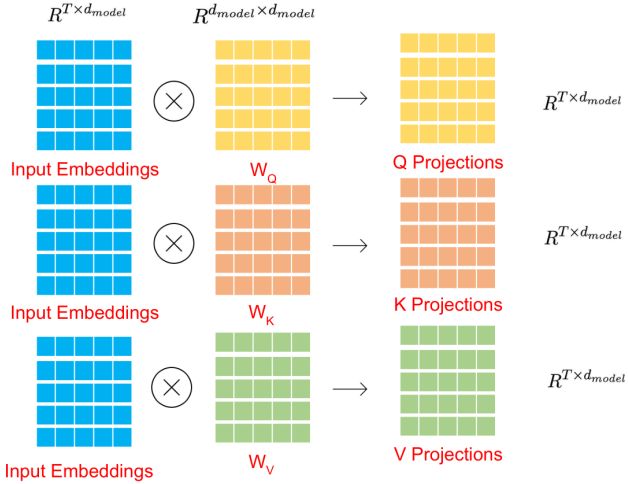
Query Inputs = Key Inputs = Value Inputs

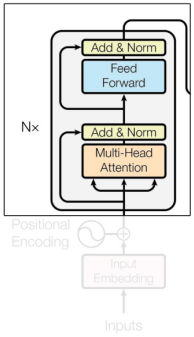
Self Attention





Self Attention





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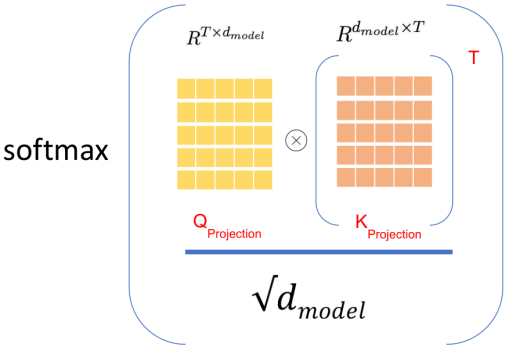
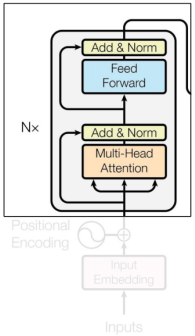
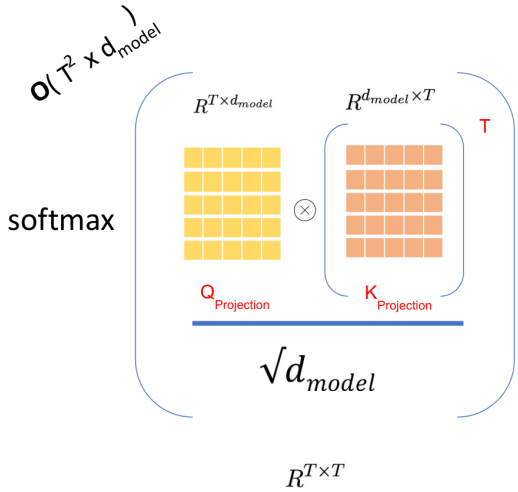
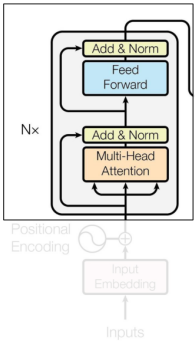


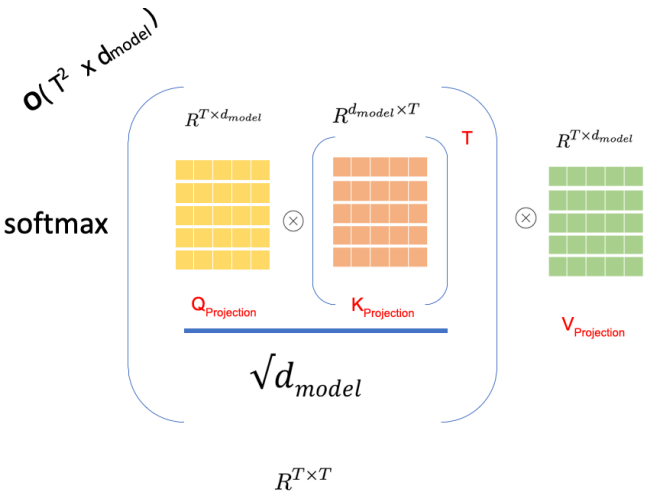


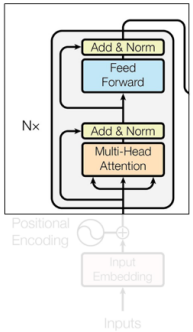
Self Attention





Self Attention

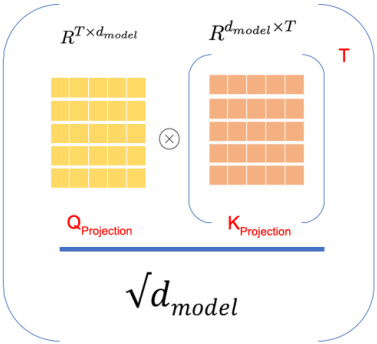




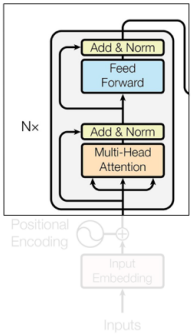
Self Attention

$o(T^2 \times d_{model})$

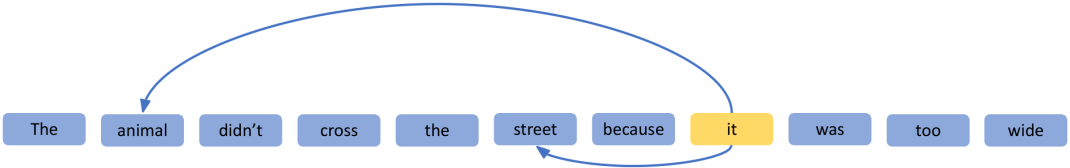
softmax



$R^{T \times T}$

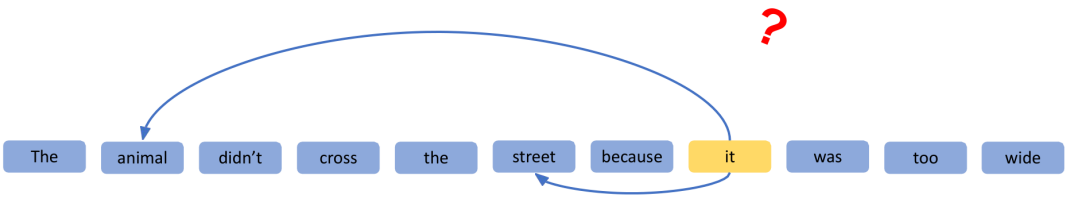


Self Attention



Coreference resolution ✓

Self Attention



Sentence boundaries ?

Coreference resolution ✓

Context ?

Semantic relationships ?

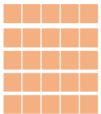
Part of Speech ?

Comparisons ?

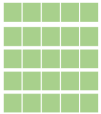
Self Attention

$$R^{d_{model} \times d_{model}}$$

W_Q

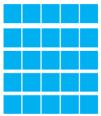


W_K

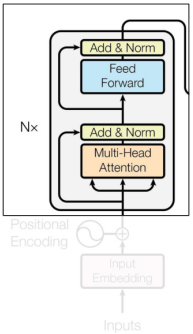


W_v

$$R^{T \times d_{model}}$$



Input Embeddings



Multi-Head Attention

- Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions.
- It consists of multiple attention heads, each with its own set of parameters.
- The outputs of all heads are concatenated and linearly transformed.

Formula:

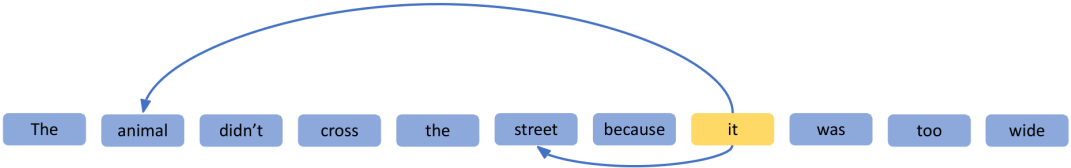
$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O$$

where each head is defined as:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

- W_i^Q, W_i^K, and W_i^V are learned projection matrices for each head.
- W^O is the output projection matrix that combines the outputs of all heads.

Multi-Head Attention

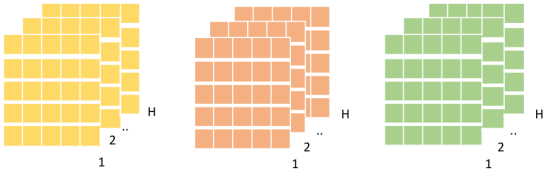


Sentence boundaries ✓ Coreference resolution ✓ Context ✓

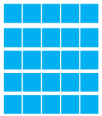
Semantic relationships ✓ Part of speech ✓ Comparisons ✓

Multi-Head Attention

$R^{d_{model} \times d_h}$



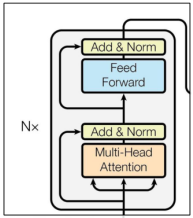
$W_{Q1,}$ $W_{Q2,\ \ldots}$ $W_{QH,}$ $W_{K1,}$ $W_{K2,\ \ldots}$ $W_{KH,}$ $W_{V1,}$ $W_{V2,\ \ldots}$ $W_{VH,}$



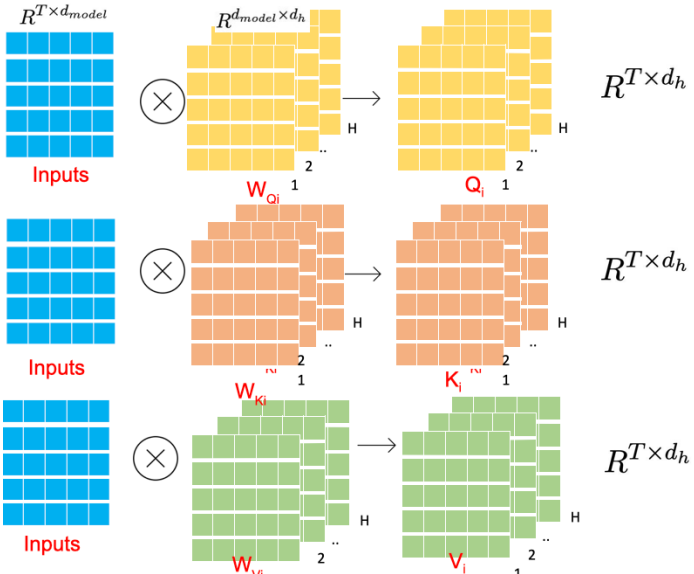
Input Embeddings

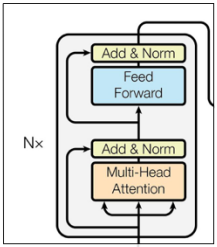
$R^{T \times d_{model}}$

$$d_h = \frac{d_{model}}{h}$$

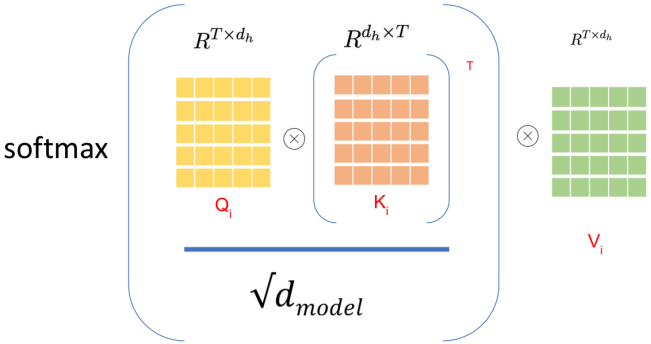


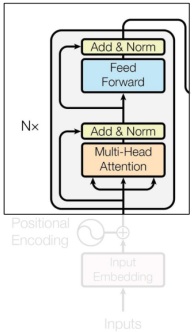
Multi-Head Attention





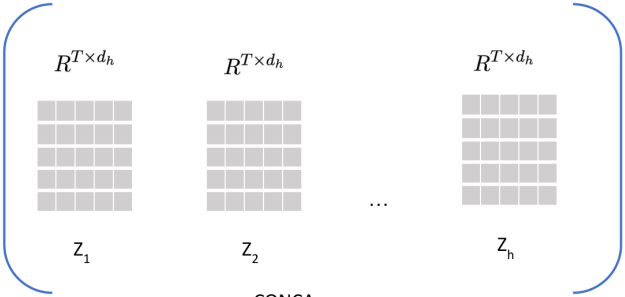
Multi-Head Attention

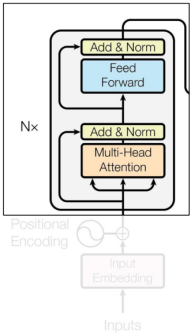




for all i ∈ [1, h]

Multi-Head Attention



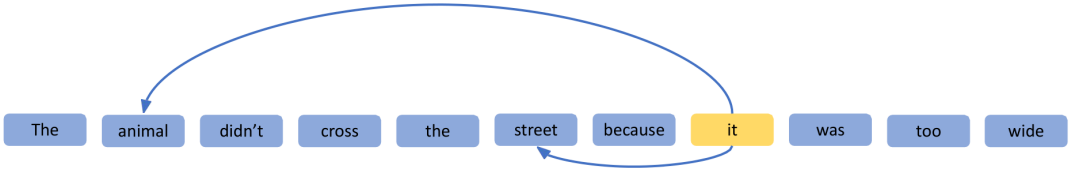


Multi Head Attention : Z

$$d_h = \frac{d_{model}}{h}$$

$$R^{T \times d_{model}}$$

Multi-Head Attention



Sentence boundaries ✓ Coreference resolution ✓ Context ✓

Semantic relationships ✓ Part of speech ✓ Comparisons ✓

Benefits of Multi-Head Attention

- Enables the model to jointly attend to information from different representation subspaces at different positions.
- Captures various linguistic and syntactic patterns by using multiple attention heads.
- Improves the model's ability to represent complex relationships in the data.
- Provides richer and more diverse feature representations.

Positional Encoding

- Transformers lack sequence order awareness.
- **Positional encodings** are added to input embeddings to provide information about token positions.
- **Sinusoidal encoding:**

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{\text{model}}}}}\right)$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{\text{model}}}}}\right)$$

Position Encoding

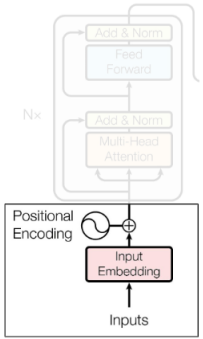
For each position, an embedded input is moved the same distance but at a different angle. **Inputs that are close to each other in the sequence have similar perturbations, but inputs that are far apart are perturbed in different directions.**

$$PE_{(pos,\,2i)} = \sin\left(pos/10000^{2i/d_{\text{model}}}\right)$$

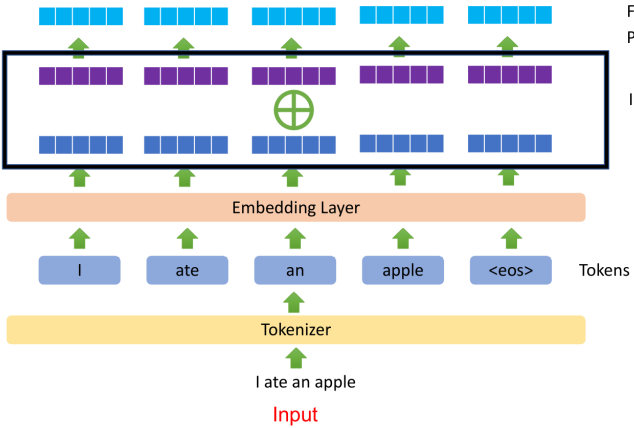
$$PE_{(pos,\,2i+1)} = \cos\left(pos/10000^{2i/d_{\text{model}}}\right)$$

pos idx of the token in input sentence
i i^{th} dimension out of d
d_{model} embedding dimension of each token

Different calculations for odd and even embedding indices.



Position Encoding



Alternatives to Positional Encoding

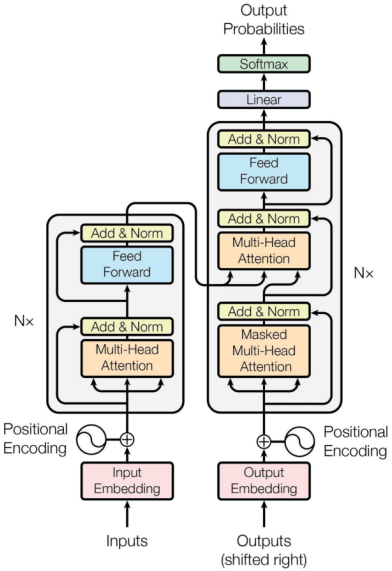
- **Learnable position embeddings:**
  - Position embeddings are parameters learned during training.
  - Allow the model to adapt position representations to the task.
- **Rotary Position Embeddings (RoPE):**
  - Encodes positions by rotating query and key vectors in multi-head attention.
  - Enables better extrapolation to longer sequences.
- **ALiBi (Attention with Linear Biases):**
  - Adds linear biases to attention scores based on distance between tokens.
  - Encourages attention to nearby tokens without explicit position embeddings.

Transformer Block

- **Multi-head Attention**
- **Add & Norm**
- **Feed-forward Layer**
- **Add & Norm**

Note
Residual connections and layer normalization are crucial for effective training.

Transformer Block



Pre-LayerNorm vs. Post-LayerNorm

Feature	Pre-LN	Post-LN
Stability	✓ More stable	✗ Less stable
Gradient flow	✓ Better	✗ Can explode/vanish
Used in	GPT-3, T5	Original Transformer

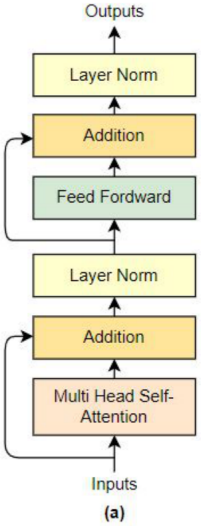
Table 1: Comparison of Pre-LayerNorm and Post-LayerNorm Architectures

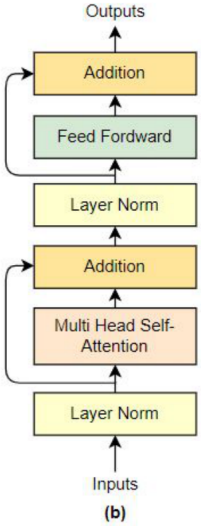
Pre-LN: LayerNorm applied **before** sublayers

Post-LN: LayerNorm applied **after** sublayers

Pre-LayerNorm vs. Post-LayerNorm

Post-LN

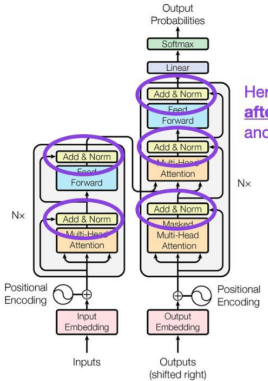




Pre-LN

Pre-LayerNorm vs. Post-LayerNorm

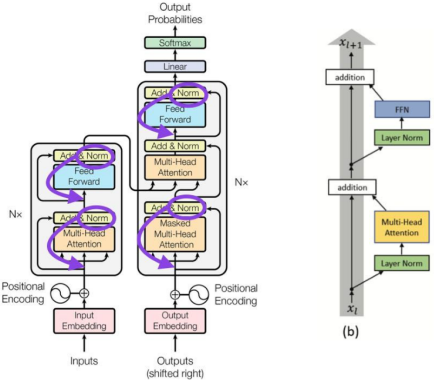
Post-LN Transformer



Here, LayerNorms are **after** attention and **after** fully connected layers

Better gradients if LayerNorm is placed inside residual connection, before the attention and fully connected layers

Pre-LN Transformer



Places the layer normalization between the residual blocks: the expected gradients of the parameters near the output layer are large

Layer Normalization vs RMSNorm

LayerNorm: Normalizes across the feature dimension by subtracting the mean and dividing by the standard deviation.

RMSNorm: Removes the mean and scales by the root mean square (RMS) of the features.

- Simpler
- Faster
- Slightly better in large-scale settings.

Key-Value Caching

- In **decode**, the model emits **one token per step**, but that token still depends on the **key and value** tensors for **all** earlier positions.
- Those positions include the **prompt** (K/V from **prefill**) plus any **new** K/V computed up to the current step.
- **KV caching** keeps these tensors in GPU memory so they are **not recomputed** at every step; each iteration **appends** the newly computed K/V to the cache for the next step.
- In many stacks there is **one KV cache per transformer layer**.

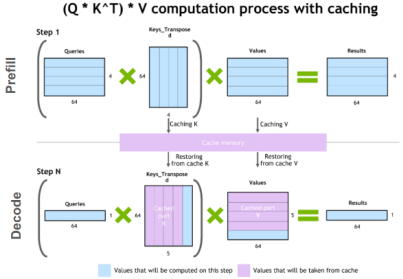


Figure 1: illustration of the key-value caching mechanism.

Verma & Vaidya, "Mastering LLM Techniques: Inference Optimization," NVIDIA Technical Blog (2023). Article — Key-value caching (Figure 1).

KV Cache Memory

- Major GPU memory uses for serving: **model weights** and the **KV cache** (self-attention tensors cached to skip redundant work).
- Under batching, each request still needs a **separate** KV-cache allocation—often the limiting factor before weights alone.
- Cost scales **linearly** with batch and length—caps throughput and long-context use; motivates MQA/GQA, paging (PagedAttention), KV quantisation, etc.

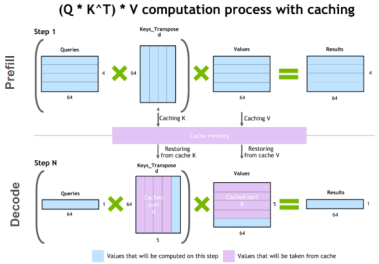
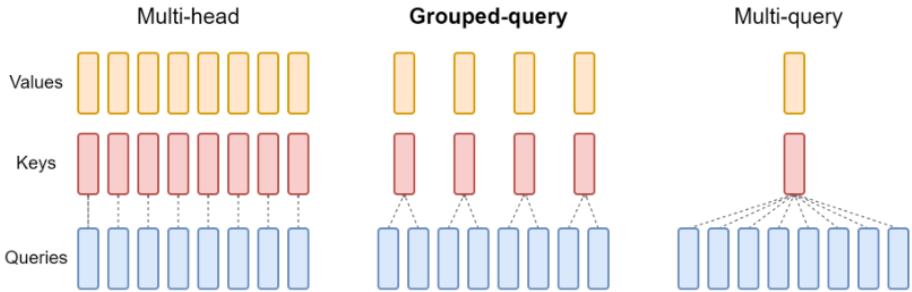


Figure 1: illustration of the key-value caching mechanism (from previous frame).

MHA, MQA, GQA: shrinking the KV cache



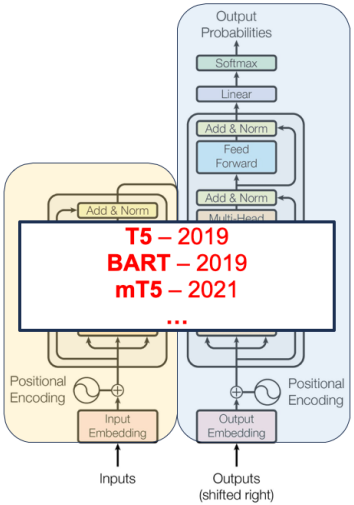
- **Multi-head attention (MHA)**: H separate query, key, and value heads.
- **Multi-query attention (MQA)**: single shared key and value heads for all query heads.
- **Grouped-query attention (GQA)**: shares key and value heads for each *group* of query heads.
- GQA conceptually interpolates between multi-head and multi-query attention.

Reference: Ainslie et al., "GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints," arXiv:2305.13245.

The LLM Era – Paradigm Shift in Machine Learning

BERT – 2018
DistilBERT – 2019
RoBERTa – 2019
ALBERT – 2019
ELECTRA – 2020
DeBERTa – 2020
…

Representation



GPT – 2018
GPT-2 – 2019
GPT-3 – 2020
GPT-Neo – 2021
GPT-3.5 (ChatGPT) – 2022
LLaMA – 2023
GPT-4 – 2023
…

Generation

Decoder-only LLMs (post-2017 trend)

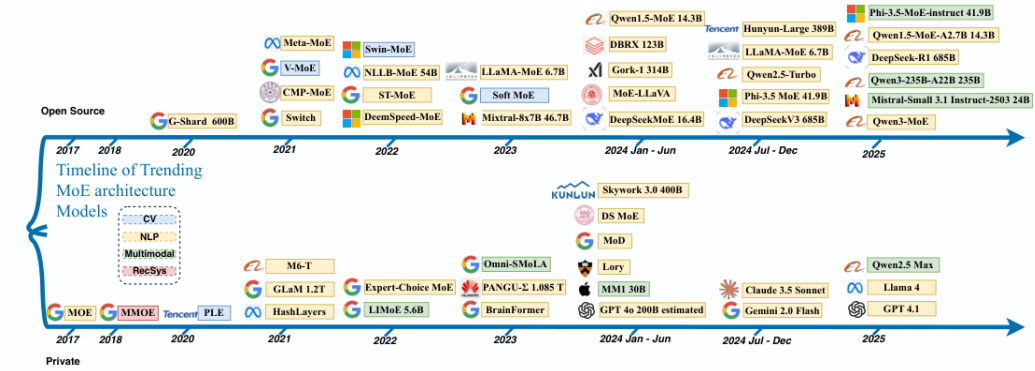
- GPT-style **causal (masked) self-attention**: each token attends only to the past.
- Single stack replaces separate encoder/decoder for many language (and vision–language) workloads.
- Encoder–decoder Transformers remain standard in **seq2seq** (MT, Whisper ASR, some VL captioning) where cross-attention helps.

Mixture of Experts (MoE)

- A dense feed-forward block always runs the same large MLP for every token. An MoE block replaces that MLP with several smaller expert MLPs.
- A lightweight router (gating network) looks at each token and picks one or a few experts to run. The rest stay idle for that token.
- Why it helps: you can increase total parameters (model capacity) without increasing active FLOPs per token as much as a fully dense model—similar spirit to "widen capacity, keep per-token work bounded."
- Routing can become unbalanced, communication-heavy in distributed training, or collapse onto a few experts unless training is regularized.

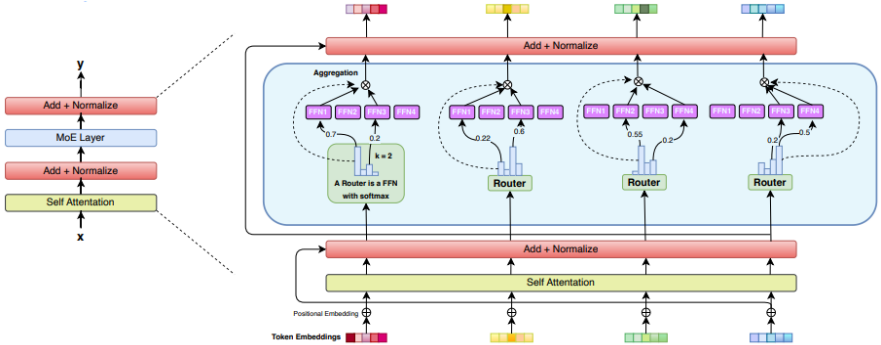
Sources: Zhang et al., "Mixture of Experts in Large Language Models," arXiv:2507.11181; Kranen & Nguyen, NVIDIA Technical Blog (Mar. 2024).

MoE milestones



Reference: Zhang et al., "Mixture of Experts in Large Language Models," arXiv:2507.11181.

MoE for a decoder-only Transformer

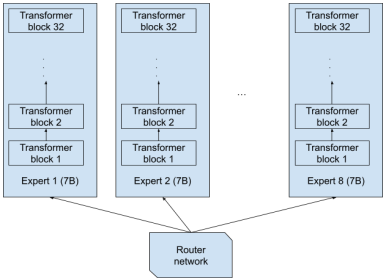


MoE for a decoder-only Transformer with $k = 2$ experts.

- Input **x** enters a **gating / router** that outputs scores for each expert.
- Only the top-k experts run their FFN; outputs are **combined** (often a weighted sum) for the residual stream.

Example: Mixtral 8×7B — a common misconception

- The name sounds like "eight separate 7B models." The **released architecture is not that**: experts are **not** eight full standalone stacks.
- Instead, think "**eight expert FFNs** per MoE layer, with **shared** attention and norms," and only a subset fires per token (here, **top-2** experts).



Reference: Kranen & Nguyen, "Applying Mixture of Experts in LLM Architectures," NVIDIA Technical Blog (2024), article.

MoE: Most Preferred Tokens By Experts

Reference: Kranen & Nguyen, "Applying Mixture of Experts in LLM Architectures," NVIDIA Technical Blog (2024), article.

Vision–Language Models: why pair vision with language?

- Strong **image embeddings** are useful, but many tasks still need a way to **say what you want** in words (“what species of bird is this?”, “read the sign in the corner”).
- **Language** makes it easy to specify new tasks without collecting a fresh classification label set for every concept—a path to flexible, user-steerable vision AI.
- **Nomenclature (informal):** an **LLM** is language-first; a **VLM** adds a **vision encoder** so the stack can **condition on images**; an **MLLM** usually means several modalities (image, video, audio) in one family.

Bandaru, *Vision Language Models* (blog, Aug. 2025), rohitbandaru.github.io/blog/Vision-Language-Models/.

Fusion Patterns: Where do image tokens meet the LLM?

- Early fusion: build one long token sequence (visual tokens + text tokens) and run attention over the mixture—what most "native" VLMs aim for today.
- Middle fusion: insert vision cross-attention inside some transformer blocks (common when adapting a legacy text-only stack such as early LLaMA 3.2).
- Late fusion: combine modalities only at the end—usually too weak for open-ended captioning/VQA because the LLM never gets rich visual state early enough.
- Images create many tokens; good designs compress visual length (pooling, grouping, tiling) before the LLM sees them.

Early Fusion approaches: MLP Adapter

- Freeze (or lightly train) a strong ViT; map its patch tokens through a small MLP projector into the LLM embedding width, then concatenate with text tokens.
- Simple, stable, and very common in open recipes (LLaVA, LLaMA 4-style early fusion, PaliGemma 2, DeepSeek-VL, . . .).

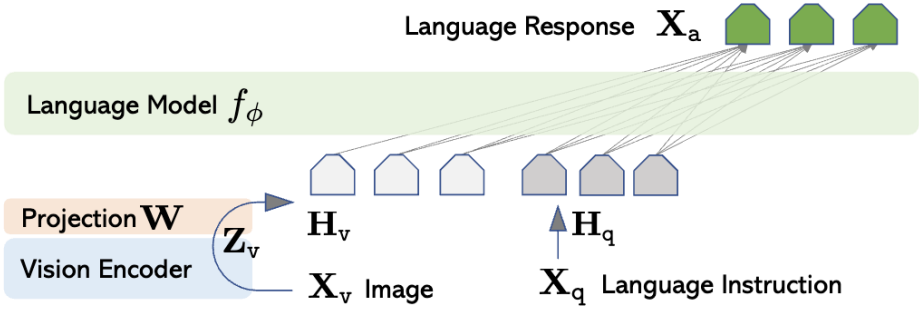


Diagram from Bandaru. Paper: Liu et al., LLaVA, arXiv:2304.08485.

Early Fusion approaches: Cross-Attention

- A small **adapter** uses learned query vectors that **cross-attend** to ViT tokens and output a **fixed small** set of visual summaries for the LLM.

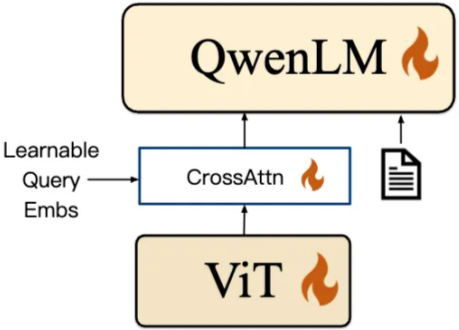


Diagram from Bandaru (Qwen-VL "position-aware" adapter).

Middle Fusion Approaches: Flamingo

- Keep a pretrained ViT and LLM frozen; insert Perceiver resamplers (compress variable tokens → fixed count) and gated cross-attention blocks inside the LLM stack.

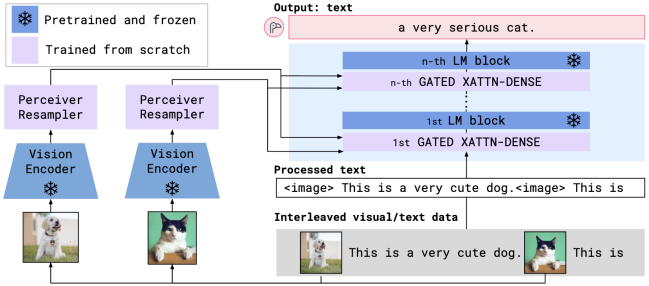


Diagram from Bandaru. Paper: Alayrac et al., Flamingo, arXiv:2204.14198.

Mid Fusion Approaches: LLaMA 3

- **ViT encoder:** 32-layer ViT trained CLIP-style for image-text alignment; features extracted from multiple layers.
- **Decoder:** Image cross attention in every fourth block (no gating on cross attention).
- **Mid fusion** suits LLaMA 3's text-first design; LLaMA 4 moves to early fusion (MLP adapter) for true multimodal training.

Summary: Transformers 2017 vs. 2026

Component	2017 Transformer	2026 consensus
Attention mechanism	Multi-Head Attention (MHA)	Grouped-Query Attention (GQA) / Multi-Head Latent Attention (MLA)
Position encoding	Absolute sinusoidal (or learned absolute)	Rotary Positional Embeddings (RoPE)
Normalization	Post-Layer Normalization (post-LN)	Pre-Layer Normalization with RMSNorm
Activation	ReLU in the FFN	SwiGLU / gated linear units
Bias terms	Biases in linear layers common	Bias-free or minimal-bias architectures
Parameter density	Dense (all parameters active per forward)	Sparse activation (e.g., Mixture-of-Experts)

Summary

- The slowest part is often getting data in and out of memory, not just the math.
- New models (2024–2026) are designed together, optimizing attention, normalization, position encoding, sparsity (MoE), and serving (memory, paging, kernels).
- The main goals: cut down on moving data, adjust compute to the input's length and type, and add more parameters without needing a lot more computation per token.

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